

Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Engg.) Final Examination-2023
EEE 2109: Electronics (3 Credits)

Time: 3 Hours

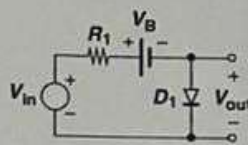
Full Marks: 70

INSTRUCTIONS

- Answer 05 out of 07 questions
- Figures in the margin indicate full marks

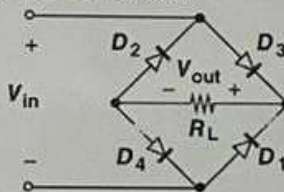
Question-1

- Explain the behavior of a p-n junction diode under reverse and forward bias conditions. 8
- Plot the input/output characteristics of the circuit shown in Fig. below using an ideal model for the diodes. 6



Question-2

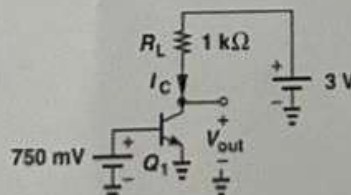
- Assuming a constant-voltage model for the diodes, plot the input/output characteristic of a full-wave rectifier circuit depicted in Fig. below. 4



- Draw the circuit diagram of a full-wave rectifier circuit and describe its operation. 6
- Given the scenario of a cell phone receiving a signal from a base station, where the signal strength increases as the user approaches the base station, there is a risk that the signal level may become large enough to saturate the circuits within the receiver chain. To prevent saturation and ensure proper functioning, it is crucial to limit the signal amplitude. Design a circuit to limit the signal amplitude to ± 100 mV as the user comes closer to the base station, with the diode turn-on voltage, $V_{D, \text{on}}$ set at 800 mV. 4

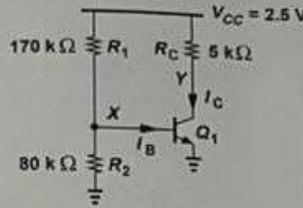
Question-3

- Describe the operation of a BJT in the active mode. Include the conditions required for the base-emitter and base-collector junctions. 7
- Determine the output voltage in Fig. below if $I_S = 5 \times 10^{-16}$ A. 7

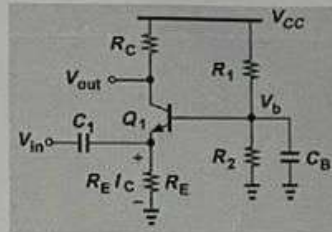


Question-4

- a. Calculate the collector current of Q_1 in the following Figure. Assume $\beta = 100$ and $I_s = 10^{-17}$ A. 6

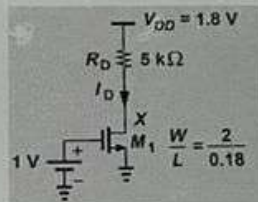


- b. Design a CB stage in the following Figure for a voltage gain of 10 and an input impedance of 50 Ω. Assume $I_s = 5 \times 10^{-16}$ A, $V_A = \infty$, $\beta = 100$, and $V_{CC} = 2.5$ V. 8



Question-5

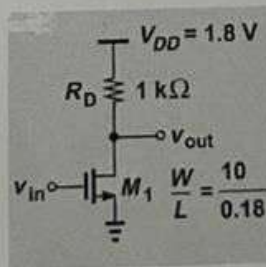
- a. Explain the channel pinch-off in MOSFET. 5
- b. Calculate the bias current of M_1 in the following Figure. Assume $\mu_n C_{ox} = 100 \mu\text{A/V}^2$ and $V_{TH} = 0.4$ V. If the gate voltage increases by 10 mV, what is the change in the drain voltage? 5



- c. Plot the I_D - V_{DS} characteristics for different values of V_{GS} for a MOSFET and briefly explain its salient features. 4

Question-6

- a. Explain the small signal model for the Common Source MOSFET. 5
- b. A MOSFET is biased at a drain current of 0.5 mA. If $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $W/L = 10$, and $\lambda = 0.1 \text{ V}^{-1}$, calculate its small-signal parameters. 3
- c. Calculate the small-signal voltage gain of the CS stage shown in the following Fig. below if $I_D = 1$ mA, $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $V_{TH} = 0.5$ V, and $\lambda = 0$. Verify that M_1 operates in saturation. 6



Question-7

- a. Explain the gain, input resistance, and output resistance of both inverting and non-inverting operational amplifier (op-amp) configurations. 8
- b. The circuit shown in Fig. below is called a "unity-gain" buffer. Note that the output is tied to the inverting input. Determine the output voltage if $V_{in} = +1$ V and $A_o = 1000$. 6

